

Heart Disease Classification

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Abstract

This project investigates the limitations of linear machine learning models for heart disease prediction using the UCI Heart Disease dataset. Several classification approaches were implemented and compared, including Logistic Regression, k-Nearest Neighbors (KNN), Support Vector Machines (SVM) with a polynomial kernel, and a simplified implementation of XGBoost. The objective was to analyze how non-linear learning methods improve predictive performance compared to linear models. Experimental results were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and model comparison plots. The study demonstrates that non-linear models are better able to capture complex relationships between clinical features and heart disease outcomes.

1 Introduction

Heart disease remains one of the leading causes of mortality worldwide. Early prediction of cardiovascular diseases can significantly improve preventive healthcare and patient outcomes. Machine learning techniques provide a data-driven approach capable of discovering hidden patterns in medical datasets and assisting clinicians in risk assessment.

The UCI Heart Disease dataset contains multiple clinical measurements such as age, cholesterol level, resting blood pressure, maximum heart rate, and electrocardiographic information. The relationship between these variables and heart disease is often complex and non-linear, making the problem suitable for the study of different machine learning approaches.

The primary objective of this project is to investigate the limitations of linear models and evaluate how non-linear methods improve classification performance. To achieve this goal, four models were compared:

- Logistic Regression (implemented from scratch)
- k-Nearest Neighbors (implemented from scratch)
- Polynomial Kernel SVM
- XGBoost with sigmoid output

The study aims to answer three main questions:

1. Why do linear models fail in certain classification scenarios?
2. How do kernel-based methods improve decision boundaries?
3. Why does boosting often achieve superior predictive performance?

2 Dataset Analysis

The experiments were conducted using the UCI Heart Disease dataset obtained from Kaggle. The dataset contains patient medical information and a binary target indicating the presence or absence of heart disease.

2.1 Dataset Characteristics

- Number of samples: 1000
- Number of features: 14
- Target variable: disease
- Missing values: 0

The target variable was transformed into a binary classification problem where positive values correspond to patients diagnosed with heart disease and negative values correspond to healthy individuals.

2.2 Class Distribution

The distribution indicates whether the dataset is balanced or imbalanced, which directly affects classifier performance.

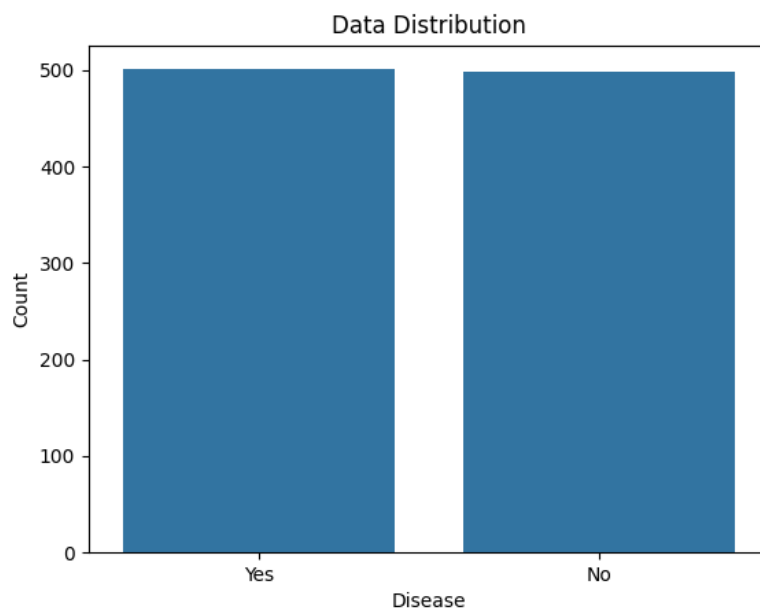


Figure 1: Distribution of heart disease classes.

2.3 Feature Correlation Analysis

To better understand the relationships between clinical variables, a correlation matrix was computed and visualized using a heatmap. Correlation analysis helps identify which features are most strongly associated with the target variable and reveals dependencies among predictor variables.

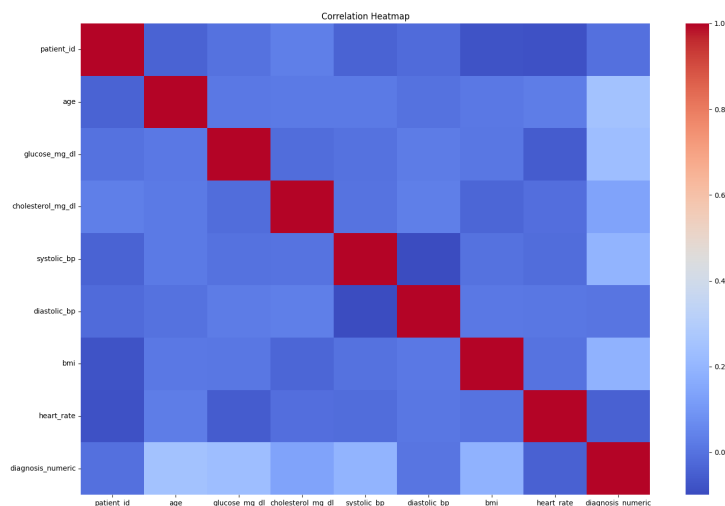


Figure 2: Correlation matrix of the Heart Disease dataset.

The heatmap highlights the degree of linear association between pairs of variables. Positive correlations indicate that two variables tend to increase together, while negative correlations indicate inverse relationships.

Features exhibiting stronger correlations with the target variable may provide greater predictive value for classification models. However, the absence of strong linear correlations does not necessarily imply a lack of predictive information, as non-linear relationships may still exist. This observation motivates the comparison between linear methods such as Logistic Regression and non-linear approaches such as SVM and XGBoost.

3 Methodology

3.1 Logistic Regression

Logistic Regression is a linear classification algorithm that models the probability of a patient belonging to a class using the sigmoid function.

$$P(y = 1|x) = \sigma(w^T x + b)$$

The model learns parameters by minimizing Binary Cross Entropy loss through gradient descent. Because the decision boundary is linear, Logistic Regression may struggle when the relationship between features and labels is non-linear.

3.2 k-Nearest Neighbors

k-Nearest Neighbors is an instance-based learning algorithm. Instead of building an explicit model, predictions are made by examining the labels of the k closest training samples.

Classification is performed through majority voting among neighbors. The method naturally captures non-linear decision boundaries but may become sensitive to noise and computationally expensive for large datasets.

3.3 Support Vector Machine

Support Vector Machines (SVMs) were introduced to overcome some of the limitations of traditional linear classifiers. While Logistic Regression attempts to find a decision boundary that separates classes by minimizing classification error, SVM focuses on finding the boundary that maximizes the distance between classes.

3.3.1 Maximum Margin Principle

Consider a binary classification problem where several separating lines are possible. SVM selects the hyperplane that maximizes the margin, defined as the minimum distance between the decision boundary and the nearest training samples.

A larger margin generally leads to better generalization on unseen data because the classifier becomes less sensitive to small perturbations in the input features.

The decision boundary is defined as:

$$w^T x + b = 0$$

where w represents the normal vector of the hyperplane and b is the bias term.

The margin is inversely proportional to the norm of w :

$$Margin = \frac{2}{||w||}$$

Therefore, maximizing the margin is equivalent to minimizing $||w||$ while correctly classifying the training samples.

3.3.2 Support Vectors

Unlike Logistic Regression, SVM does not use all training examples equally. Only the observations closest to the decision boundary influence the final solution. These critical points are called *support vectors*.

The position of the support vectors completely determines the separating hyperplane. Points located far from the boundary have little or no influence on the final classifier. This property makes SVM particularly robust to irrelevant observations and noise.

3.3.3 Soft-Margin SVM

Real-world datasets are rarely perfectly separable. In medical datasets, patients with similar clinical measurements may belong to different classes, making perfect separation impossible.

To address this issue, SVM introduces slack variables that allow certain samples to violate the margin constraints. The optimization problem becomes:

$$\min_{w,b} \frac{1}{2} ||w||^2 + C \sum_{i=1}^n \xi_i$$

where:

- $||w||^2$ controls model complexity,
- ξ_i are slack variables,
- C controls the trade-off between maximizing the margin and reducing classification errors.

Large values of C prioritize correct classification, whereas smaller values allow more violations in exchange for a larger margin.

3.3.4 Kernel Trick

Although maximizing the margin improves classification, many datasets are not linearly separable in the original feature space.

The kernel trick solves this problem by implicitly mapping the data into a higher-dimensional feature space where a linear separation may become possible. Instead of explicitly computing the transformed features, SVM computes a kernel function:

$$K(x, z) = \phi(x)^T \phi(z)$$

which measures similarity between observations in the transformed space.

This approach allows the algorithm to learn highly non-linear decision boundaries without explicitly constructing high-dimensional feature vectors.

3.3.5 Polynomial Kernel

In this project, a polynomial kernel was implemented:

$$K(x, z) = (x^T z + c)^d$$

where:

- d controls the polynomial degree,
- c is a constant offset parameter.

Polynomial kernels implicitly generate higher-order feature interactions while avoiding the computational cost of explicit feature transformation. The kernel therefore enables SVM to model complex relationships between clinical variables such as age, cholesterol, blood pressure, and maximum heart rate.

This capability is particularly important for heart disease prediction, where relationships between risk factors are rarely purely linear.

3.4 XGBoost

XGBoost (Extreme Gradient Boosting) is an ensemble learning algorithm based on the principle of gradient boosting. Unlike traditional machine learning models that learn a single decision boundary, XGBoost combines multiple weak decision trees to construct a stronger predictive model.

3.4.1 Ensemble Learning

The fundamental idea behind ensemble learning is that a collection of weak models can often achieve better predictive performance than a single complex model. In the case of XGBoost, each individual tree is intentionally kept simple and acts as a weak learner. The final prediction is obtained by combining the outputs of all trees in the ensemble.

Rather than building all trees independently, XGBoost constructs them sequentially. Each new tree focuses on correcting the mistakes made by the previous trees.

3.4.2 Gradient Boosting Principle

Suppose an initial model produces predictions that contain errors. Instead of rebuilding the entire model, gradient boosting trains a new tree to learn these residual errors.

If the current model prediction is denoted by

$$F_m(x),$$

the next tree is trained to approximate the negative gradient of the loss function, which indicates the direction of greatest improvement.

The updated model becomes

$$F_{m+1}(x) = F_m(x) + T_m(x),$$

where $T_m(x)$ represents the newly added tree.

By repeatedly correcting previous mistakes, the ensemble gradually improves its predictive performance.

3.4.3 Binary Classification with Sigmoid Output

For heart disease prediction, the task is binary classification. Following the implementation presented in the Kaggle notebook *Building XGBoost From Scratch*, the accumulated tree outputs are converted into probabilities using the sigmoid activation function.

$$P(y = 1) = \sigma(F(x)) = \frac{1}{1 + e^{-F(x)}}$$

where $F(x)$ is the sum of the outputs produced by all trees.

The resulting probability is then compared to a decision threshold to assign the final class label.

3.4.4 Loss Function

The model optimizes Binary Cross Entropy (BCE) loss:

$$L(y, \hat{y}) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

where y represents the true label and $\hat{y}$ is the predicted probability.

The gradients and Hessians of this loss function are used to determine how new trees should be constructed in order to reduce prediction error.

3.4.5 Tree Splitting Strategy

A key idea of XGBoost is that each decision tree is built by selecting splits that maximize the reduction of the loss function. Candidate splits are evaluated using gradient statistics collected from the training samples.

The gain associated with a split measures how much the split improves the objective function. Splits producing larger gains are preferred because they lead to greater reductions in prediction error.

3.4.6 Why XGBoost is Non-Linear

Unlike Logistic Regression, which learns a single linear decision boundary, XGBoost partitions the feature space into multiple regions through decision trees. Each tree captures different feature interactions, and the ensemble combines these interactions into a highly flexible model.

For heart disease prediction, risk factors such as age, cholesterol level, blood pressure, chest pain type, and maximum heart rate often interact in a complex manner. These interactions are difficult to represent using a single linear equation but can be effectively captured by boosted decision trees.

Consequently, XGBoost is capable of modeling highly non-linear relationships and often achieves superior predictive performance compared to linear classification methods.

4 Experimental Setup

4.1 Data Preprocessing

The dataset was cleaned and transformed before training. Labels were encoded into binary classes. For the XGBoost implementation, labels were converted from $\{-1,1\}$ to $\{0,1\}$ to match the requirements of binary cross-entropy optimization.

4.2 Train-Test Split

The dataset was divided into training and testing subsets to evaluate model generalization on unseen samples.

4.3 Evaluation Metrics

The following metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score

5 Results

5.1 Performance Comparison

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.8650	0.8696	0.8421	0.8556
KNN	0.7900	0.8118	0.7263	0.7667
Polynomial SVM	0.8550	0.9125	0.7684	0.8343
XGBoost	0.9400	0.9462	0.9263	0.9362

Table 1: Performance comparison of all models.

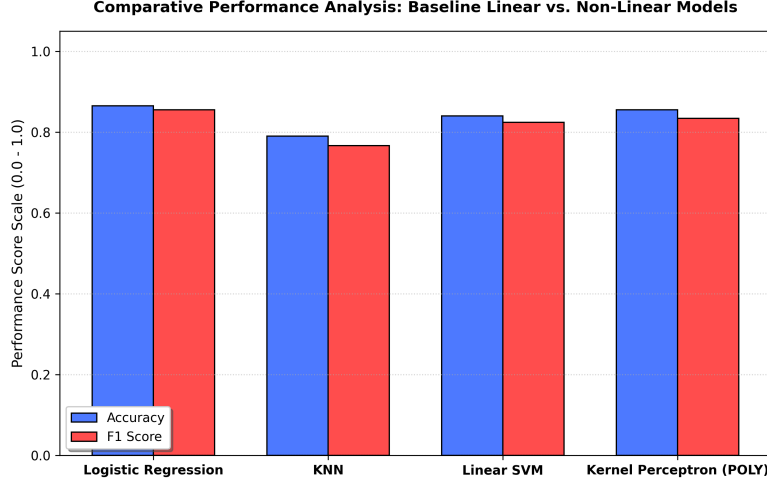


Figure 3: Accuracy comparison among all evaluated models.

6 Discussion

Experimental results reveal the limitations of linear decision boundaries in heart disease prediction. Logistic Regression assumes a linear separation between healthy and diseased patients, which may not adequately capture the complex interactions among clinical variables.

KNN improved performance by exploiting local neighborhood information, allowing non-linear decision boundaries. The polynomial kernel SVM further improved classification by projecting the data into a higher-dimensional feature space while maximizing the classification margin.

The strongest performance was achieved by XGBoost. Its boosting strategy iteratively corrected previous mistakes and successfully modeled complex feature interactions. These results support the hypothesis that non-linear methods provide superior predictive power for heart disease prediction.

7 Limitations

Several limitations should be considered.

- The dataset size is relatively small compared to modern medical datasets.
- Clinical measurements may contain noise and measurement errors.
- Hyperparameter tuning was limited due to computational constraints.

- The custom implementation of XGBoost represents a simplified version of the industrial algorithm.
- Results are specific to the selected train-test split.

8 Conclusion

This project investigated the limitations of linear models for heart disease prediction and evaluated several non-linear alternatives.

The experiments demonstrated that Logistic Regression provides a useful baseline but struggles to capture complex relationships among clinical variables. KNN and SVM improved performance through local decision making and feature transformations, while XGBoost achieved the strongest results through ensemble learning and gradient boosting.

The study experimentally confirms that non-linear methods are often more effective for medical classification problems where decision boundaries are complex. Future work could investigate larger datasets, advanced boosting techniques, and deeper hyperparameter optimization.